Name _____ Period _____ Date _____

Rusty Stripes

Focus question: What are examples of how Earth has changed and how life has changed?

Warm-up

Iron rusts when it meets oxygen. What does the rust in a banded iron formation tell you about the air or water that touched it long ago? Write two questions.

Pre-check

About how old is Earth? Name one way Earth has changed. Name one way life has changed. Did living things ever change the air or the rocks?

Card sort

Glue in: Card sort chart and timeline strip

Name _____ Period _____ Date _____

Rusty Stripes (continued)

Focus question: What are examples of how Earth has changed and how life has changed?

Unit CER evidence log

Glue in: Unit CER evidence log (or start it in the back of your notebook)

Preview the unit

Star the target you think will be hardest.

Exit ticket

Name one way Earth changed and one way life changed.

Name _____ Period _____ Date _____

Rusty Stripes: Reading Notes

Focus question: How did Earth and life evolve together?

Warm-up

Skim the headings. What do you predict?

Reading: Lesson 8.3 Earth and Life Evolve Together (pp. 226-235)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (coevolution, extinction, fossil record)

Name _____ Period _____ Date _____

Rusty Stripes: Reading Notes (continued)

Focus question: How did Earth and life evolve together?

Main idea in my words

Question I still have

Connect it

Sketch one example from the reading of Earth changing life or life changing Earth.

Exit ticket

One new idea from the reading is ____.

Name _____ Period _____ Date _____

A Two-Way Street

Focus question: How did Earth change life, and how did life change Earth?

Warm-up

An ice sheet creeps across a region. What happens to plants and animals? A lichen grows on bare rock for many years. What might happen to the rock?

Sort

Glue in: Sorting chart and diagrams

Find it in the reading

Copy one sentence that shows life changing Earth and write the page number. Add one row to your evidence log.

Name _____ Period _____ Date _____

A Two-Way Street (continued)

Focus question: How did Earth change life, and how did life change Earth?

Draw your own pair

One E->L and one L->E: two boxes joined by an arrow. Under each write "First, ... so ...".

Change goes in ___ directions

Exit ticket

One way Earth changed life: ____. One way life changed Earth: ____.

Name _____ Period _____ Date _____

A Two-Way Street: Reading Notes

Focus question: How did early life change Earth?

Warm-up

What do you know about early life?

Reading: Lesson 8.2 Life's Early History (pp. 219-225)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (photosynthesis, stromatolite, oxygen)

Name _____ Period _____ Date _____

A Two-Way Street: Reading Notes (continued)

Focus question: How did early life change Earth?

Main idea in my words

Question I still have

Connect it

Sketch one example from the reading of Earth changing life or life changing Earth.

Exit ticket

Early life changed Earth by ____.

Name _____ Period _____ Date _____

The Oxygen Story

Focus question: How can one event in Earth's history change life, and how can life change Earth in return?

Warm-up

Rust needs oxygen. What do you predict about oxygen in the air and ocean when the rust stripes formed?

Graph

Glue in: Oxygen story: data table and graph grid

The rust record

Which came first: oxygen from living things, or the rust layers?

Name _____ Period _____ Date _____

The Oxygen Story (continued)

Focus question: How can one event in Earth's history change life, and how can life change Earth in return?

Mass extinctions

Glue in: Mass extinction data and the rust record

Build a chain

Glue in: Build a chain

Wrap-up

Because _____ happened, _____ changed, which led to _____.

Exit ticket

Because ____ happened, ____ changed, which led to ____.

Name _____ Period _____ Date _____

Three Case Files

Focus question: How do events in Earth's history change life, and how does life change Earth in return?

Warm-up

Look at your chain from yesterday. Which arrow shows life changing Earth? Which shows Earth changing life? Tell a partner.

Case files

Glue in: Three case files

Name _____ Period _____ Date _____

Three Case Files (continued)

Focus question: How do events in Earth's history change life, and how does life change Earth in return?

What do all three cases have in common?

Quick write

Pick one case. Write 3 sentences with the word because that tell how Earth changed life and how life changed Earth.

Exit ticket

Choose the case you understand best and describe the loop in one sentence.

Name _____ Period _____ Date _____

Make the Argument

Focus question: How do I write an argument with a claim, evidence, and reasoning?

Warm-up

A friend says, "Earth changed life, but life never changed Earth." Do you agree? One sentence and one reason.

Evidence bank

Glue in: Evidence bank cards you chose (or list E-numbers) and add three rows to your evidence log.

Plan

Glue in: Argument planner and peer-check scale

Name _____ Period _____ Date _____

Make the Argument (continued)

Focus question: How do I write an argument with a claim, evidence, and reasoning?

Write your argument

One paragraph of 8 to 10 sentences. Use at least three pieces of evidence, include a number, name the kind, use because twice.

Peer check

Score: ___ Star: ___ Step: ___

Exit ticket

One thing my partner suggested that I will fix: ____.

Name _____ Period _____ Date _____

Evidence Court

Focus question: How do different kinds of evidence work together to make one strong argument?

Warm-up

Would you believe a story told by only one witness? What would make you trust it more? Write 2 sentences.

Sort and rate

Glue in: Rating table and argument frames

Why do different kinds make a claim stronger?

Name _____ Period _____ Date _____

Evidence Court (continued)

Focus question: How do different kinds of evidence work together to make one strong argument?

Build one argument

Write the paragraph using the frames.

Depth challenge (choose two)

Different perspectives; Over time; Unanswered questions; Big idea.

Exit ticket

The strongest piece of evidence in my table is ____ because ____.

Name _____ Period _____ Date _____

Earth's Systems Web

Focus question: How are Earth's systems and life connected over time?

Warm-up

Write the four Earth systems and one example in each.

Warm-up systems

Geosphere (rock): ... Hydrosphere (water): ... Atmosphere (air): ... Biosphere (living things): ...

Systems web

Glue in: Earth's systems web template and reading chart

Name _____ Period _____ Date _____

Earth's Systems Web (continued)

Focus question: How are Earth's systems and life connected over time?

Sketchnote

Make a sketchnote of your web with color, picture, text, and a template (about equal pictures and text).

Which system has the most arrows out? In?

Exit ticket

Which system has the most arrows pointing out of it? Which has the most pointing in?

Name _____ Period _____ Date _____

Earth's Systems Web: Reading Notes

Focus question: How did Earth form and change?

Warm-up

Skim the headings. What system is each about?

Reading: Lesson 8.1 Earth's Origin (pp. 206-218)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (geosphere, hydrosphere, atmosphere, biosphere)

Name _____ Period _____ Date _____

Earth's Systems Web: Reading Notes (continued)

Focus question: How did Earth form and change?

Main idea in my words

Question I still have

Connect it

Sketch one example from the reading of Earth changing life or life changing Earth.

Exit ticket

One new idea about early Earth is ____.

Name _____ Period _____ Date _____

Argue Together

Focus question: Can I argue from evidence that Earth's systems and life changed together over time?

Warm-up

Look at your web from Lesson 7. Pick the arrow you think matters most. Why?

The timeline

Glue in: The timeline: eight rows

Name _____ Period _____ Date _____

Argue Together (continued)

Focus question: Can I argue from evidence that Earth's systems and life changed together over time?

Write your argument

One paragraph (8 to 10 sentences).

Share and fix

Check: claim both ways, three rows with numbers, because. One thing I fixed: ____

Exit ticket

One loop in Earth's history is ____.

Name _____ Period _____ Date _____

Review and Reteach

Focus question: What do I still need to practice for the CSA?

Warm-up

List your six CFA scores (ESS2-7.1 to 7.6). Circle any under 4 of 5 correct.

My CFA scores

List ESS2-7.1 to 7.6. Circle any under 4 of 5 correct (3 of 4 points).

Review work

Glue in: Review packet A to C

Name _____ Period _____ Date _____

Review and Reteach (continued)

Focus question: What do I still need to practice for the CSA?

Review work

Glue in: Review packet D to F.

Can you...?

Put a check next to each target you are sure of.

Exit ticket

Tomorrow is the CSA. The target I will study tonight is ____.

Name _____ Period _____ Date _____

CSA

Focus question: Can I show I have met the targets for this unit?

Warm-up

Before the CSA, I feel ____ about ____.

What I learned about Earth and life changing together

Three things I understand better now.

The rusty stripes one more time

Explain the banded iron formation with Earth and life changing each other.

Exit ticket

The one idea I want to remember from this unit is ____.